SEMESTER II 2025/2026 SECD2613-04

SYSTEM ANALYSIS AND DESIGN

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GROUP PROJECT PHASE 2 - GROUP 2

GROUP PROJECT : UTM TRANSPORT APP

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1.0 Overview of the Project

Universiti Teknologi Malaysia, as one of the best technological universities in the world, has many students who move around every day from their residences to the various faculties. Therefore, efficient campus transportation is no longer just a convenience but a vital necessity. The university operates an internal shuttle bus service running different routes within the university campus connecting various residential colleges, faculties, and administration blocks.

However, the current bus service is not satisfactory to the students due to several reasons such as the lack of active buses with high student density. These issues lead to the inconvenience when using the bus service provided and may affect the daily study plan such as students being unable to attend the class on time due to the delay of the buses.

Thus, this project is to design and develop a new UTM Transport App to ensure that students can get transportation easily on campus. This mobile application focuses on bringing the physical and virtual sides of campus life together through the use of real-time data analysis and mobile applications to maximize efficiency and engagement within the campus ecosystem.

In this project, we will collect the students' suggestions or even complaints, and analyse the pain points of the existing transport system. After that, the existing features in the current system will be modified and the new features will be created based on students' requirements. All of these features will be included in our UTM Transport App. Therefore, instead of switching various mobile applications, students can get satisfactory transportation by using only a single mobile application.

2.0 Problem Statement

At a campus as huge as Universiti Teknologi Malaysia (UTM), transportation is not just a "nice-to-have" feature, it is the heartbeat of our daily academic lives. However, the current transportation system tends to be fragmented and traditional that just cannot keep up with the intense schedule of a modern student. For most of us, arranging our traveling schedule is not the smooth transition it should be. It has become a daily source of stress and friction that drains our focus and productivity. To alleviate this issue, we need to address five specific "pain points" that are currently making life harder for the UTM community:

1. **The Bus Arrival Uncertainty**

   Right now, the biggest headache is the total lack of transparency. We are forced to stand at bus stops for who-knows-how-long, often sweating under the scorching sun or getting soaked in a heavy rain, without any idea if the bus is two minutes away or twenty. This uncertainty causes a domino effect, such that we end up late for crucial lectures, miss the start of lab sessions, and live in a constant state of "transit anxiety" that makes it impossible to plan our study groups or even a simple lunch break effectively.

2. **The Peak-Hour Seat Shortages**

   During the morning and evening rush, the campus transport system basically turns into "survival of the fittest" mode. This is because we have no way to check how full a bus is in advance, we often have to watch bus after bus drive past because they are packed with people from the previous stations. It is not just about the delay, but also about the discomfort of being squeezed into overcrowded vehicles and the safety risks that come with it, leaving those at later stops completely stranded.

3. The Inefficient Carpooling

So many of us are willing to share rides or land a helping hand to ease the load on the buses, but there is not a safe, official way to make that happen. We are currently stuck using disorganized social media chats or word-of-mouth, which is awkward and unreliable. This "fragmented carpooling" means thousands of seats go empty every day while other students are left waiting at the station or walking with their bare feet, wasting a huge opportunity to build a better campus community and lower our carbon footprint.

4. No Route Planning Tools

UTM is huge, and for "freshies" or visitors, it can feel like a total maze. Without a smart tool to help plan a route, students struggle to find the fastest way to get from their residential colleges to distant faculties or admin buildings. We end up wasting time taking long, inefficient paths or walking much further than we need to, simply because we do not have a map that shows us how to navigate the transit transfers correctly.

5. The Lack of Feedback

The most frustrating part of all this is that when things go wrong, we have no real voice. There is not a dedicated, systematic way to report things like a bus being consistently late, a vehicle being dirty, or safety concerns with driving. Without this feedback, the university admin is essentially "flying blind," making it impossible for them to see where the service is failing or how to fix it. This silence leads to a cycle of the same problems happening semester after semester with no end in sight.

3.0 Proposed Solutions

- **Design a Secure User Management Module**
  This module focuses on account management functions such as account creation, authentication, and user-profile maintenance to ensure that users of the application are indeed verified UTM students. They need to sign up using UTM email. So, users just need to register their account and click on the "Remember Me" checkbox if one wants quick future logins.

- **Implement a Live Tracking & Mapping Module**
  This module offers users the ability to view live information on campus buses' routes and locations. The information displayed consists of the actual location of the bus on the map, along with an Estimated Time of Arrival (ETA) at their current station. The module enables users to pick any bus on the map and observe its movement, along with its plate number.

- **Implement a Seat Availability Monitor Module**
  This module allows users to view real-time seat availability of approaching campus buses. The information displayed includes the current occupancy level (available, moderate, or full). So, users can select a bus and check its seat status along with its route, helping them decide whether to board or wait for the next bus.

- **Create a Carpool Module**
  This module helps in making travelling on the campus flexible since it offers the students the ability to form rides with other students. Users have two options in this module either to form their own carpool ride or join others. Users can go through the list of available rides and find someone who matches.

- **Design a Route Planning Module**

  This module acts as the smart guide in planning how users can travel on the campus in the best way. The fastest route options will be provided together with step by step instructions as well as any required bus transfer along the way. Users can simply input the destination point from the point of departure to access various route options.

- **Implement a Feedback & Rating Module**

  This module allows the users to help in improving campus transport through feedback. Here, users can rate their experience riding the bus and report problems such as lateness and overcrowding among others. Once the trip ends, a rating window pops out where users can rate the bus.

4.0 Information Gathering Process

In order to obtain a comprehensive understanding of the present state of the UTM campus transportation system, we consider both perspectives of those using it as well as through observation of the physical surroundings. Therefore, three techniques have been performed for data gathering purposes, which include Joint Application Design (JAD) session, an online questionnaire using Google Form, and Structured Observation of the Environment (STROBE).

The selection of the techniques cover both interactive and unobtrusive approaches. The JAD session and questionnaire allows stakeholders and users to directly express their past experience and provide constructive opinions, while STROBE allows our team to observe the actual transport behaviour in the UTM campus without influencing anything.

4.1 Method Used

4.1.1 Joint Application Design (JAD)

JAD refers to a structured workshop, which involves bringing in the users, stakeholders, and developers into a collaborative environment to discuss the requirements of the system. Unlike individual one-to-one interviews, JAD ensures that all interested parties are involved in one conversation to discuss their needs and expectations for the new system, as well as identifying any existing problems that must be addressed. Hence, a JAD session is appropriate because it allows our team to obtain the perspectives of different stakeholders simultaneously, reduce misunderstandings in the early process, and produce a richer set of requirements in a shorter time compared to conducting separate individual interviews.

JAD Session Details :

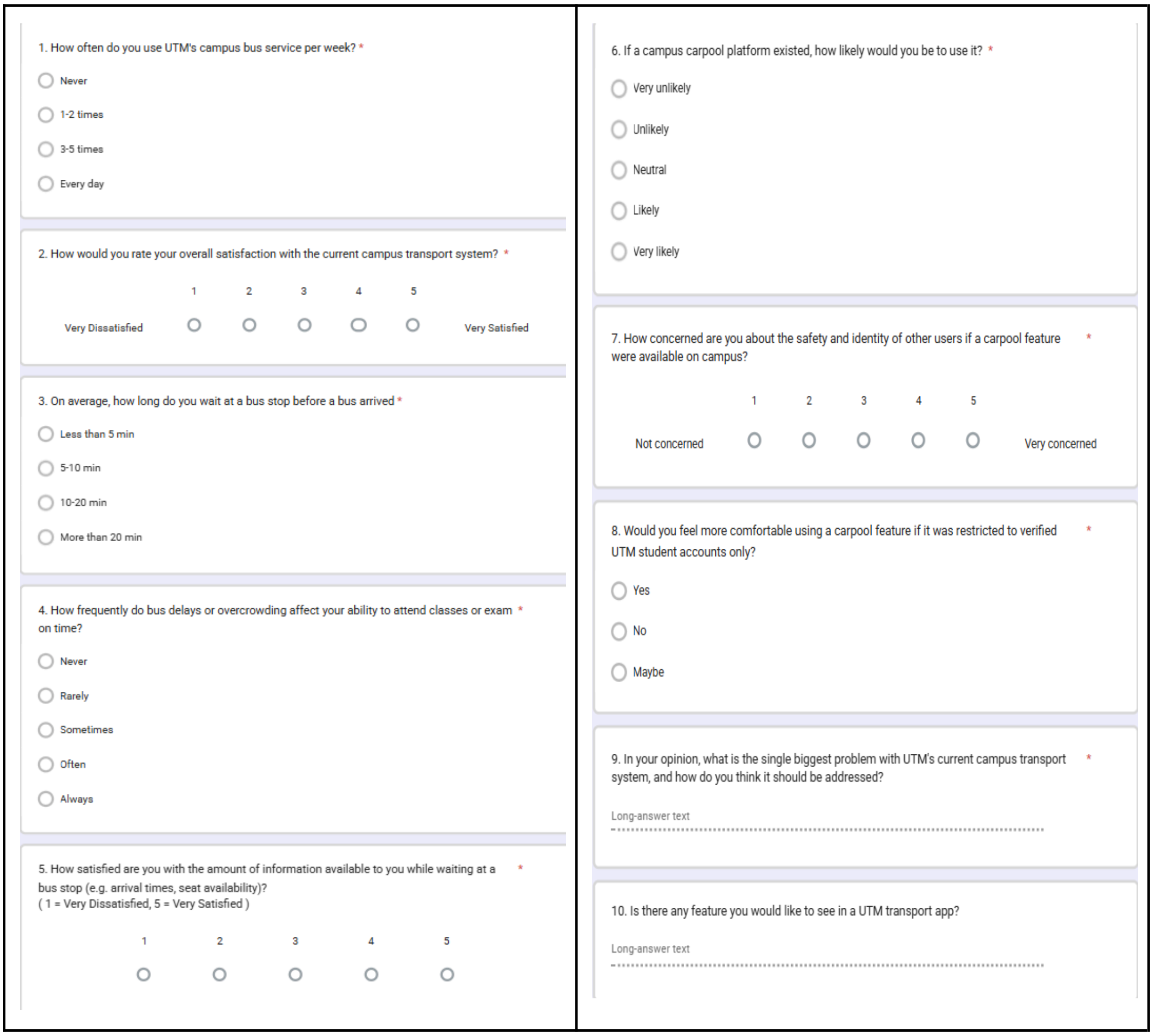
Item	Details
Participants	UTM students (representing bus users, carpool users, freshmen)
Session Leader	Chew Jian Hui, Darren Chua Boon Yee
Duration	Approximately 45 minutes
Location	Faculty of Computing, UTM JB
Topic Covered	System scope presentation, validation of proposed modules, user expectations, safety and privacy concerns

Key Discussion Questions that Raised during the JAD session:

1. Based on what was presented, do you think this system would actually solve the transport problems you face daily?
2. What information would you need from the app before deciding whether to wait for or skip a bus?
3. Have you ever used informal carpooling on campus? What made you uncomfortable about it?
4. When navigating to an unfamiliar location on campus, what kind of guidance would you expect from an app?
5. If something went wrong during your journey, how would you want to report it?
6. Are there any concerns you have about using a campus transport app, particularly around privacy or safety?

4.1.2 Questionnaire

An online questionnaire in the form of a Google Form was created to collect quantitative data among a larger group of UTM students. This questionnaire was designed on a pyramid structure, where the questions start with closed questions then expand by allowing open-ended questions and more generalized responses. We opt for pyramid structure to design the questionnaire because respondents will not be bombarded right away with complicated or even difficult questions to answer. Rather, they can comfortably answer basic and easier questions first like "How often do you take the UTM buses?" and later proceed to more general questions like "Is there any feature you would like to see in the UTM Transport App ?"



1. How often do you use UTM's campus bus service per week? *
- Never
- 1-2 times
- 3-5 times
- Every day

2. How would you rate your overall satisfaction with the current campus transport system? *

Very Dissatisfied 1 2 3 4 5 Very Satisfied

3. On average, how long do you wait at a bus stop before a bus arrived *
- Less than 5 min
- 5-10 min
- 10-20 min
- More than 20 min

4. How frequently do bus delays or overcrowding affect your ability to attend classes or exam on time? *
- Never
- Rarely
- Sometimes
- Often
- Always

5. How satisfied are you with the amount of information available to you while waiting at a bus stop (e.g. arrival times, seat availability)? *
(1 = Very Dissatisfied, 5 = Very Satisfied)

1 2 3 4 5

6. If a campus carpool platform existed, how likely would you be to use it? *
- Very unlikely
- Unlikely
- Neutral
- Likely
- Very likely

7. How concerned are you about the safety and identity of other users if a carpool feature were available on campus? *

Not concerned 1 2 3 4 5 Very concerned

8. Would you feel more comfortable using a carpool feature if it was restricted to verified UTM student accounts only? *
- Yes
- No
- Maybe

9. In your opinion, what is the single biggest problem with UTM's current campus transport system, and how do you think it should be addressed? *

Long-answer text

10. Is there any feature you would like to see in a UTM transport app?

Long-answer text

Figure 4.1.2.1 Questionnaire questions

4.1.3 Structured Observation of the Environment (STROBE)

STROBE is an unobtrusive method, which allows us to observe the user behaviour and how UTM transport works without directly interacting with or influencing anything. Rather than just merely relying on what users report about their experience, STROBE successfully captures what actually happened and provides a first-hand view of the current system shortcomings. In fact, the students may sometimes fail to report an accurate past travel experience during the JAD session or questionnaire. Such failure could result in underestimating the waiting time experienced by the students or failing to articulate the physical conditions they endure at bus stops on a daily basis. In that case, STROBE comes in handy since our team can witness firsthand the true nature of the bus stops, number of students, and behaviour during peak and non-peak periods.

Observation Items	Details
Average number of students waiting at bus stop	Counted at 5-minute intervals
Bus occupancy upon arrival	Empty / Available / Moderate / Full
Number of students unable to board	Counted per bus
Carpool activity in Telegram group	Any students observed arranging rides via phone while waiting
Bus schedule in UTM Smart	The way it shows the bus arrival time and destination

4.2 Summary from Method Used

4.2.1 JAD Session Summary

At the beginning of the session, an overview of the system scope was presented, which gave participants enough context to provide focused and meaningful feedback rather than just simply complaining about the situation.

Participants agreed that the new system will address their issues, but they pointed out that it will greatly depend on the reliability of information that will be displayed in the application. A recurring point was that since the current system only provides a fixed timetable with no real-time updates, students have no way of knowing whether a bus is running on time or how full it already is. Many participants emphasized the severity of this issue, because they frequently missed the morning classes or were left stranded as full buses drove past them without stopping. Participants stressed that for the new app to be trusted, its live data must be consistently accurate, otherwise it would be no more reliable than the static schedule they already have.

On the question of what information they needed before deciding whether to wait for a bus, participants consistently highlighted three things, which are current bus location, estimated arrival time, and current occupancy level. Some participants mentioned that the occupancy level was even more important than the time of arrival.

The carpooling discussion showed that most participants were familiar with the Student Grab via Telegram group but felt uncomfortable and awkward using it due to the lack of identity verification. They agreed that contact details should only be shared after both parties confirm a match, so that restricting the platform to verified UTM accounts was a necessary safety measure rather than just a nice-to-have.

Freshmen participants raised the route navigation issue most strongly, pointing out their difficulties in understanding which bus lines to use and where to make transfers when visiting unfamiliar places of campus because UTM Smart's current approach of redirecting to Google Maps was unhelpful as it does not consider any bus routes within the campus network at all. They agreed that a useful route planner must incorporate available bus routes and transfer points into its suggestions.

Regarding the feedback mechanism, participants expressed great dissatisfaction with the lack of official ways to report repeated issues, such as bus lateness and misconduct by drivers and these same problems had persisted for many semesters without noticeable actions from the university admin. To solve this, instead of a lengthy and tedious feedback form that most students would typically ignore, participants were supportive of a post-trip rating prompt with categorised complaint options, because most students would not bother with a lengthy feedback form but would readily tap a category and a star rating.

4.2.2 Questionnaire Summary

The Google Form questionnaire received 30 responses from UTM students across various faculties and years of study. The results strongly corroborate the findings from the JAD session and provide quantitative backing for the proposed system features.

Key findings from the questionnaire:

1. High bus dependency

   Over 93% of respondents reported using campus buses at least 3 times per week, confirming that bus transport is an important part of their daily life.

2. Low satisfaction

   Most students think that the current transport system did not meet their expectations and lead to inconvenience.

3. Long waiting time

   Data shows that students regularly wait around 10 to 20 minutes at a bus stop to make sure they can take the bus and prevent crowds to make sure they can have a seat.

4. Carpool interest and safety concern

   Most students are willing to try the carpool features if a proper platform existed, and they will feel more comfortable if the feature was limited to verified UTM student accounts.

5. Current problem and expectation of new app

   Most respondents stated the lack of real-time bus arrival information as their biggest frustration, because they had no way of knowing when the next bus would arrive or whether it would already be too full to board. In terms of expectations, the most commonly requested features were live bus location on a map, accurate estimated time of arrival, and real-time occupancy levels.

4.2.3 STROBE Summary

We successfully conducted the direct observation in several UTM bus stops and finally obtained the following findings.

During the morning peak hours, especially 7:15 am until 8:00 am, average 20 to 25 students were observed waiting at the bus stop at any given 5 minute interval. Among the buses that arrived during the observation period, approximately 60% arrived already at moderate-to-full capacity, leaving a significant number of waiting students unable to board. Many students were seen to board only after waiting for several consecutive buses, with waiting times exceeding 25 minutes in some cases. The same bus route was consistently observed arriving full across multiple consecutive arrivals.

Moreover, although UTM offers a bus schedule feature in UTM Smart where students can choose any bus and see its complete timetable, which shows the departure time, starting point, and the final point of the bus, this level of information is far from being enough for practical usage. However, the estimated time for intermediate stations are completely missing from the schedule, leaving only a vague time frame within which the bus should pass the stop. Additionally, there are no intelligent searches offered to the users, which means that it is impossible to find the best bus according to any particular point of interest such as a certain time or destination. In UTM Smart, we also observed that there is a feature to show some common locations in the UTM campus, but it just directly uses the Google map to show the route without specifying the way to utilize the free buses provided by UTM to go there.

Besides that, we also observed that there exists an informal carpooling activity via the Telegram group. While no dedicated platform exists for this purpose, it is evident that students have organically formed their own workaround to coordinate rides among themselves. However, this approach remains fundamentally limited, as there is no formal verification of driver identities, no structured matching system, and no accountability mechanism, leaving both drivers and passengers without adequate safety assurance.

5.0 Requirement Analysis

5.1 Current Business Process

Currently, navigating the huge Universiti Teknologi Malaysia (UTM) campus relies heavily on traditional, manual coordination and fixed timetable shuttle buses. When a student needs to travel from their residential college to a faculty building, they walk to the nearest designated bus stop and wait. Due to there is no live tracking visibility into the transit network, the student must rely on guesswork, often standing under the scorching hot sun or getting soaked during sudden heavy rains, unsure if the bus is two minutes or twenty minutes away.

During peak morning and evening hours, the situation deteriorates. When a bus finally arrives, the student frequently discovers it is already packed with passengers from the previous stops. Unable to board, the student is forced to wait for the next uncertain bus, resulting in a domino effect of missed lectures, late lab sessions, and high “transit anxiety”. If a student decides to seek alternative transport, they must resort to fragmented, unofficial carpooling. They open disorganized social media groups or messaging apps (like Telegram) to manually ask for rides, which is inefficient and poses security concerns as driver identities are not formally verified.

Furthermore, for freshmen or visitors navigating the campus maze, there is no centralized tool to plan their journey. They often take long, inefficient walking paths or miss necessary bus transfers. Finally, when the journey concludes, if the students experience a delayed schedule, reckless driving, or a dirty vehicle, there is no dedicated channel to report it. The administration remains unaware of these daily operational failures, and the cycle of frustration continues into the next semester.

5.2 Functional Requirement

User Requirement Definition

A user requirement definition was conducted to explain the services and operational constraints of the system from the perspective of the UTM students and administration, written in natural language for the stakeholders.

1. The system shall strictly authenticate users using their official UTM email addresses to ensure a secure, student-only environment.
2. The system shall display the exact real-time locations and Estimated Times of Arrival (ETA) for all campus buses on an interactive map.
3. The system shall allow users to monitor the current seat availability of the approaching buses before they arrive at the station.
4. The system shall enable students to securely host or join carpool rides with peers heading in the same direction.
5. The system shall provide an intelligent route planning feature that calculates the fastest path and guides users through campus transit transfers.
6. The system shall allow users to submit transparent feedback, rate their trips, and report specific issues immediately after their journey.

System Requirement Specification

The system requirement specification details the precise functions, services, and operational logic required to build the platform, ensuring clear guidelines for developers and stakeholders.

Live Tracking & Mapping Requirements

1.1 The system shall establish a continuous data connection with GPS transponders installed on all campus buses.

1.2 The system shall process incoming GPS coordinates and plot the actual location of each bus on the digital campus map in real time.

1.3 The system shall calculate precise Estimated Times of Arrival (ETA) based on current bus speed, distance to the user's selected station, and traffic conditions.

1.4 The system shall display specific bus details, such as the plate number and assigned route, when a user selects a bus icon on the map.

Seat Availability Monitoring Requirements

2.1 The system shall dynamically update and store the current occupancy level of each active campus bus.

2.2 The system shall categorize and visually display the real-time seat status to the user using clear indicators (e.g., Available, Moderate, or Full).

2.3 The system shall automatically prompt administrators if a specific route consistently reports a “Full” capacity over a designated time period.

Carpool Management Requirements

3.1 The system shall provide separate interfaces for “Drivers” to post available seats and “Passengers” to search for rides.

3.2 The system shall match drivers and passengers based on matching departure times and destination proximity.

3.3 The system shall facilitate secure communication by displaying verified contact details only after a ride match is confirmed by both parties.

3.4 The system shall record a digital log of all carpool transactions, including the identities of the driver and passenger, for campus safety purposes.

Intelligent Route Planning Requirements

4.1 The system shall provide an input field for users to enter their current departure point and desired campus destination.

4.2 The system shall calculate the most efficient route by factoring in walking distance, active bus route, and potential transfer points.

4.3 The system shall generate step-by-step navigation instructions for the user to follow.

Feedback and Reporting Requirements

5.1 The system shall automatically trigger a rating prompt (e.g., 1 to 5 stars) on the user’s device once a journey is marked as completed.

5.2 The system shall categorize submitted complaints (e.g., “Late Arrival”, “Vehicle Condition”, “Safety Concern”) and store them in the centralized database.

5.3 The system shall generate monthly performance reports summarizing student feedback for the university’s transport administration.

5.3 Non-functional Requirement

Non-functional requirements dictate how the system should perform, ensuring a seamless, reliable, and secure experience for thousands of concurrent UTM students. These focus on the platform's quality attributes and operational constraints. These focus on the platform's quality attributes and operational constraints.

Product Requirements

- ➔ Performance: The system shall be capable of processing live GPS data and rendering it on the user's map within 3 seconds of user interaction, avoiding any lag that could result in missed buses.
- ➔ Reliability: The system shall maintain 99.5% uptime during peak campus operating hours (Monday to Friday, 7:00 AM to 7:00 PM) to ensure students are never stranded without transit data.
- ➔ Size and Scalability: The system's cloud backend (AWS/Firebase) shall automatically scale its storage (Mbytes) and processing resources to seamlessly support up to 10000 concurrent users without performance degradation.
- ➔ Portability: The platform shall support a wide number of target systems, remaining fully operable on both iOS and Android mobile operating systems.
- ➔ Ease of Use: The user interface shall be designed with a "frictionless: approach, minimizing the training time required for a student to navigate the app and view bus ETAs.
- ➔ Robustness: The database shall have automated backup protocols. In the event of a system failure, the time to restart after failure shall not exceed 10 minutes, and the probability of data corruption on failure shall be kept below 1%.

Organizational and Process Requirements

- ➔ Process Mandates: The system development shall mandate the use of specific Integrated Development Environment (IDE) and cross-platform programming languages (e.g., Flutter) to streamline internal campus IT workflows.
- ➔ Administration: The IT team of the university's Transportation Department should be granted a secure, dedicated admin dashboard with its own unique authentication credentials to monitor network health and view feedback reports.
- ➔ Training: The system shall operate automatically from the drivers' perspective, requiring zero training time for the campus bus drivers.

External Requirements

- ➔ Integration: The system shall reliably integrate with the external Google Maps API to fetch accurate campus topographies and calculate route distances.
- ➔ Security and Privacy: The system shall implement strict data encryption (e.g., AES-256) to protect user profiles, real-time location data, and carpool logs, complying with external privacy provisions such as the Personal Data Protection Act (PDPA).
- ➔ Authentication: Users of the system shall authenticate themselves by interfacing with UTM's official domain registry to ensure that only verified emails ending in @graduate.utm.my or @utm.my can successfully register.

5.4 Logical DFD AS-IS System

<u>Context Diagram</u>

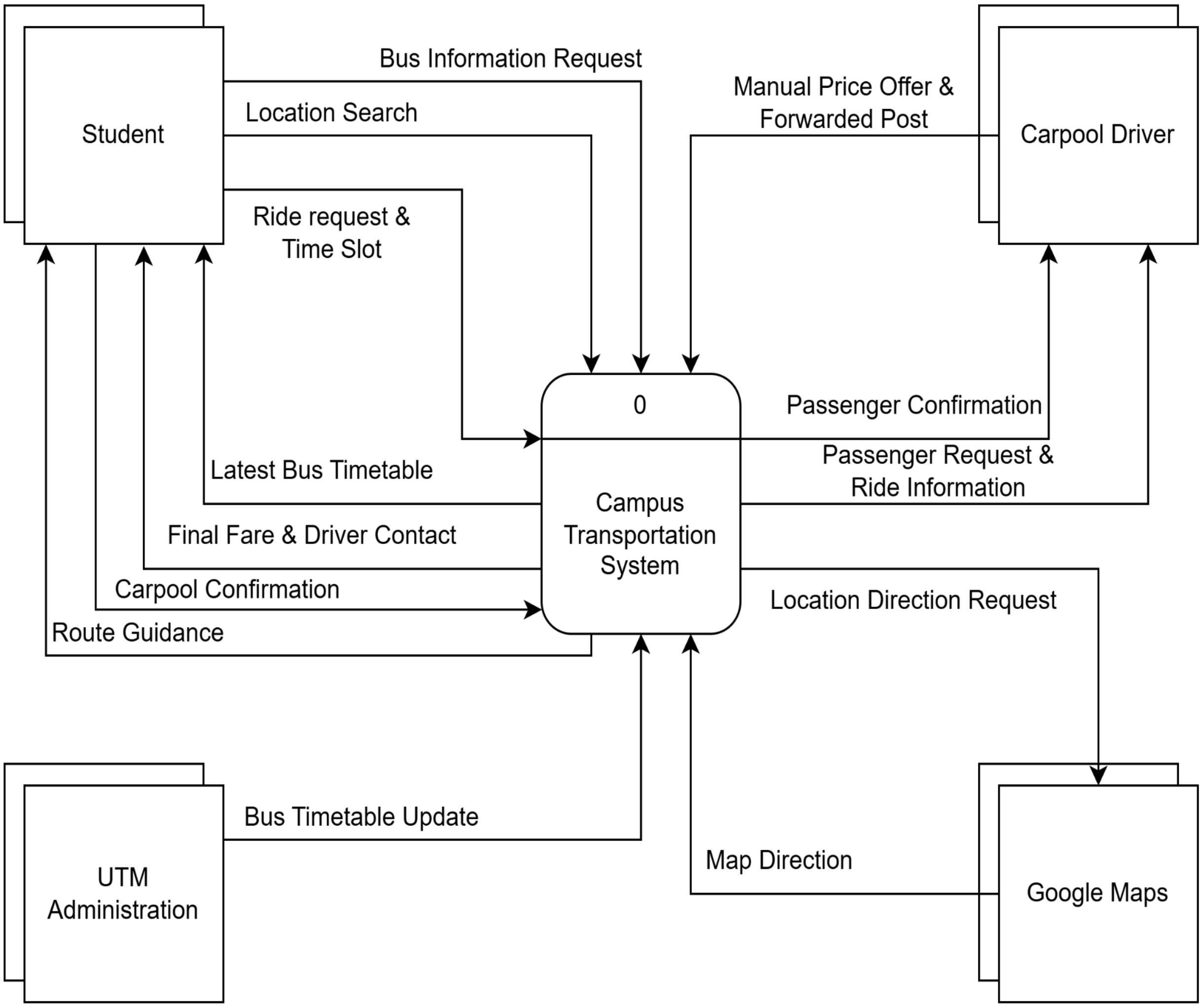
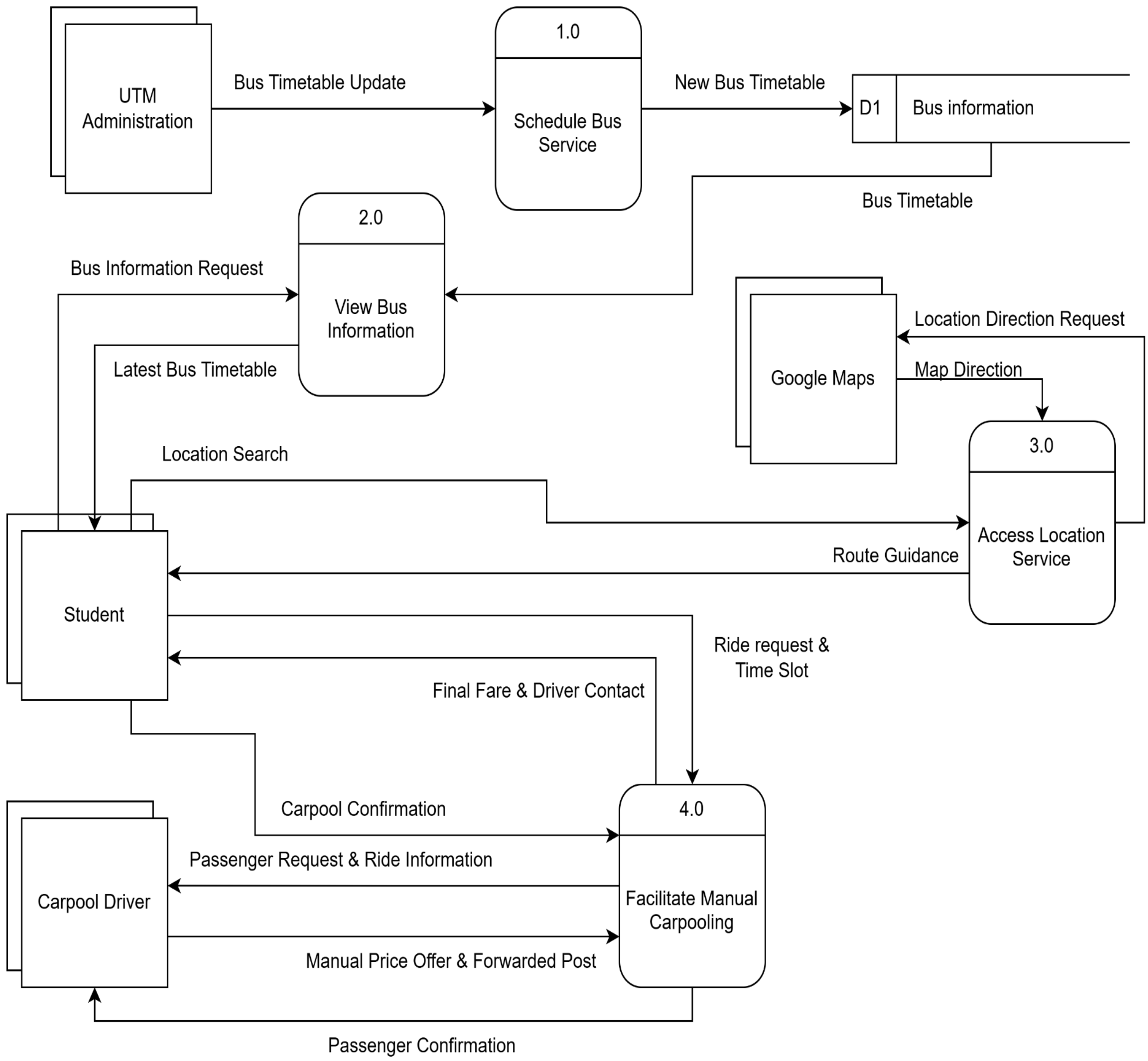


Diagram 0


UTM Administration
Bus Timetable Update
1.0
Schedule Bus Service
New Bus Timetable
D1
Bus information
Bus Timetable
2.0
View Bus Information
Bus Information Request
Latest Bus Timetable
Google Maps
Location Direction Request
Map Direction
3.0
Access Location Service
Location Search
Route Guidance
Student
Ride request & Time Slot
Final Fare & Driver Contact
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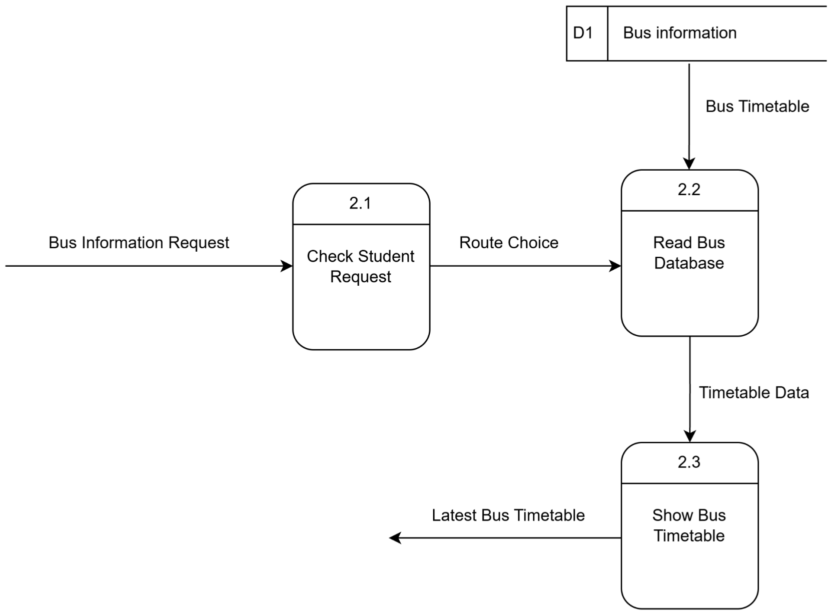
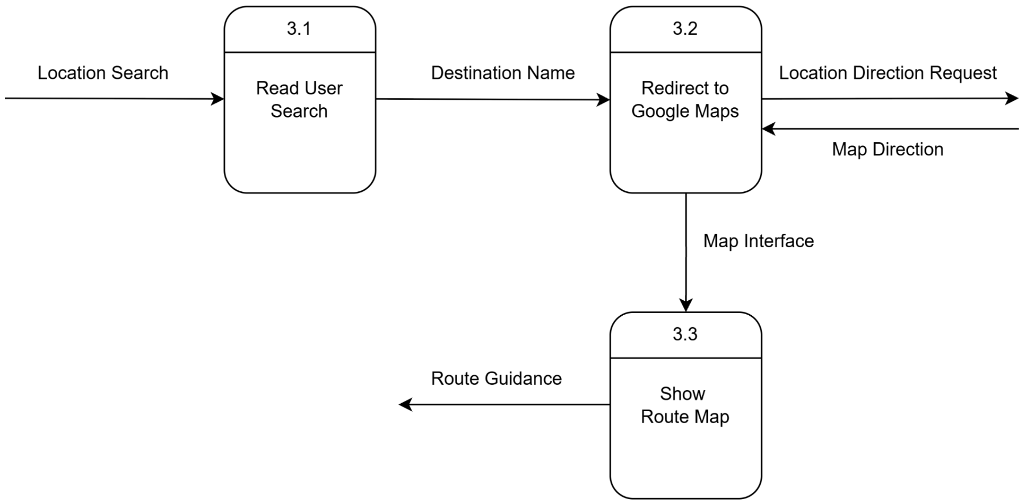
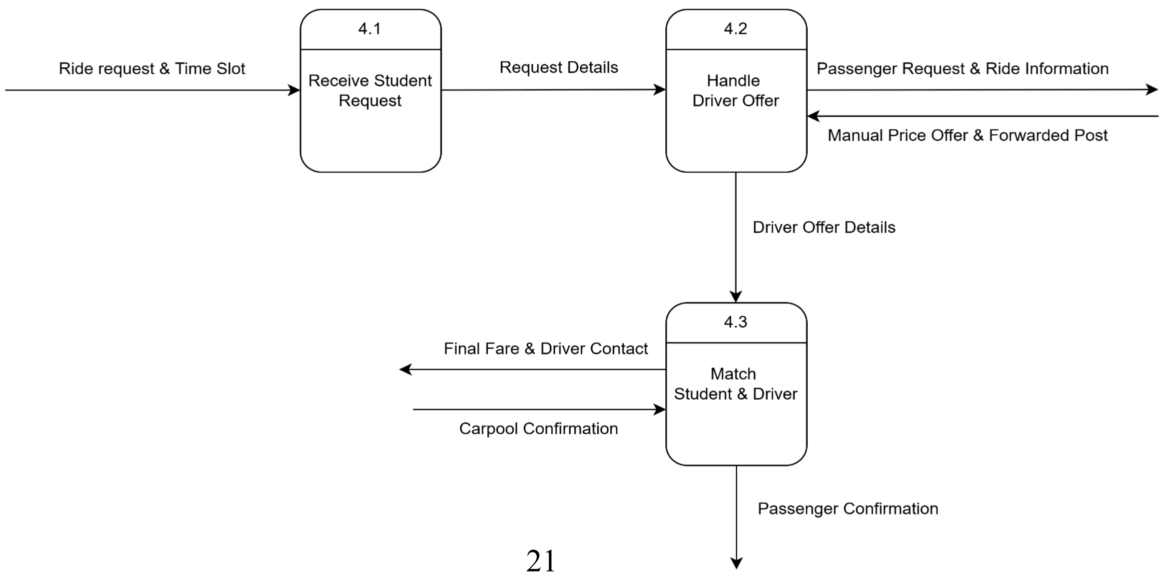
4.0
Facilitate Manual Carpooling
Carpool Driver
Passenger Request & Ride Information
Manual Price Offer & Forwarded Post
Passenger Confirmation

Child Diagram







6.0 Summary of Requirement Analysis Process

In order to propose an ideal transport system for all UTM students, we involved three information gathering processes which are Joint Application Design (JAD), Questionnaire, and Structured Observation of the Environment (STROBE). From these processes, we noticed that students are highly dependent on the buses on campus. However, the current transport system is not satisfactory to the students and we identified several problems from these three processes.

Firstly, the insufficient real time information of live bus locations and occupancy levels leads to the waste of students' time and the disruption of the plan. Therefore, a Live Tracking & Mapping feature is needed to provide the bus details and the Estimated Times of Arrival (ETA) for all campus buses. Besides that, a Seat Availability Monitoring feature is also required to process the real time occupancy level of each campus bus to ensure students can modify their plan in time if the bus is full.

Next, when students need to go somewhere they are not familiar with, the current system only provides a link to the Google map and there is no guide for students to go to the destination using the free buses provided by UTM. Thus, an Intelligent Route Planning is created to provide a guide and help students in planning the most efficient route by using the free buses to minimize the fee.

Moreover, the students showed dissatisfaction about the lack of official ways to report repeated issues causing the existing issues such as bus lateness to happen repeatedly. In order to reduce those issues, a Feedback and Reporting feature is essential which allows students to rate and submit complaints so that the administration can address those issues effectively.

Furthermore, the current transport system only provides bus services and there is no alternative transport for students to choose. This has led to the creation and use of informal social media groups among students to get transport easily. However, these informal groups are less reliable and may bring risk to both passengers and drivers. Therefore, in order to meet the students’ requirements, a Carpool Management is involved in our new transport system to provide another choice for their journey.

There are also several non-functional requirements that make the system operate and ensure the convenience of all UTM members when using it. For instance, the system shall operate automatically from drivers' perspective so that the long-time training sessions for the bus and carpool drivers can be avoided. Besides that, the user interface design focuses on the ease of use to ensure the system can be widely accepted and used by all UTM students as passengers.

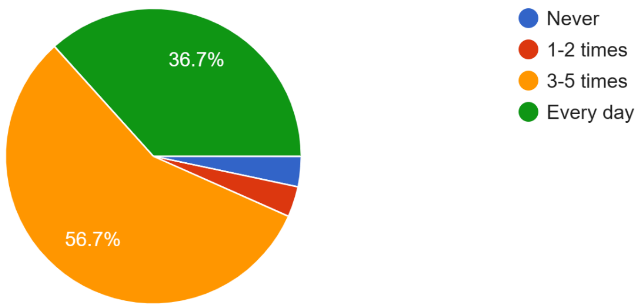
From the security perspective, our whole system will require authorization using UTM's official email during the login process. The security level of our transportation can be increased when all the bus drivers, the carpool drivers, and the passengers verify their identity using UTM's official email. Moreover, administration requires unique authentication credentials to monitor network health and view feedback reports. Lastly, our system will also implement strict data encryption to protect the privacy of data such as the user profiles and real-time location data.

In conclusion, we have collected and analysed the problems that occur in the current UTM transport system. In order to solve them, we have analysed the functional and non-functional requirements and proposed the appropriate solutions to increase the satisfaction of users. Furthermore, logical data flow diagrams of the current system were drawn to increase our understanding of the current system and make the upgrading process easier to implement.

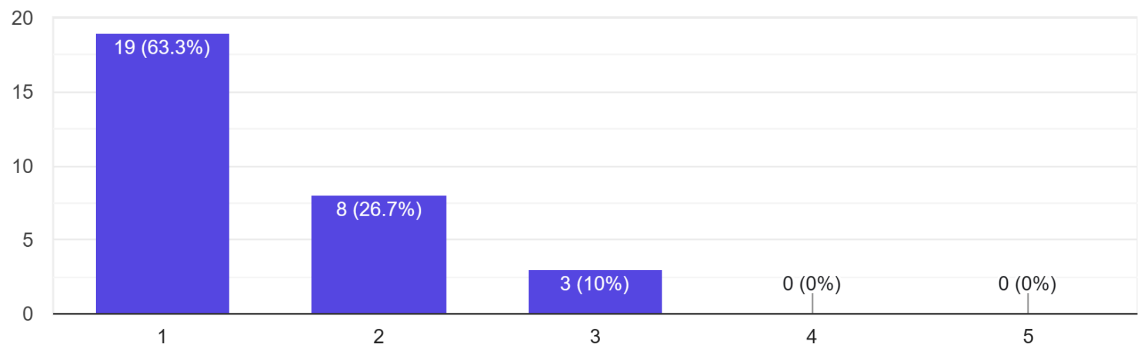
7.0 Appendix

Questionnaire responses:

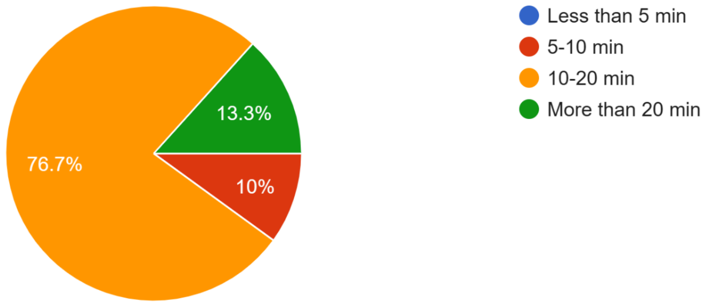
1. How often do you use UTM's campus bus service per week?
（30 条回复）



2. How would you rate your overall satisfaction with the current campus transport system?
30 responses

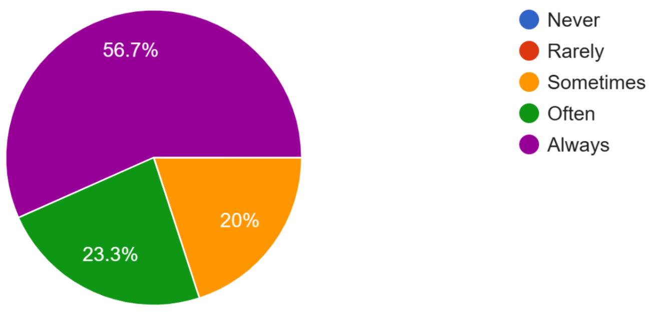


3. On average, how long do you wait at a bus stop before a bus arrived
30 responses



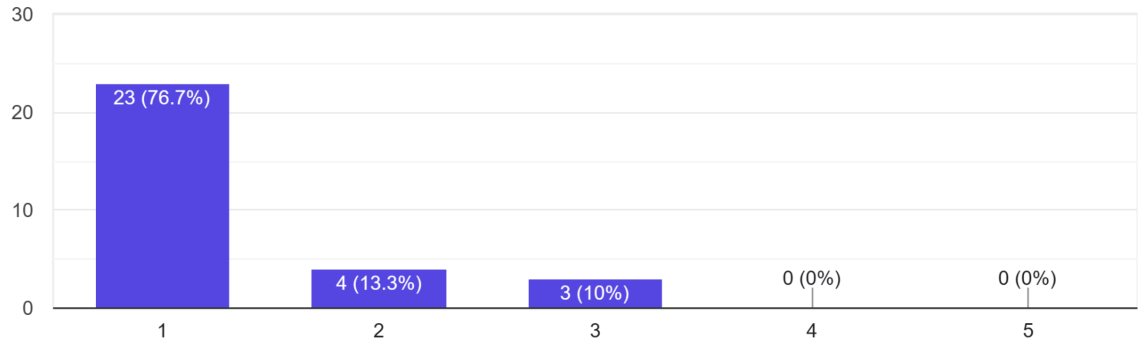
4. How frequently do bus delays or overcrowding affect your ability to attend classes or exam on time?

30 responses



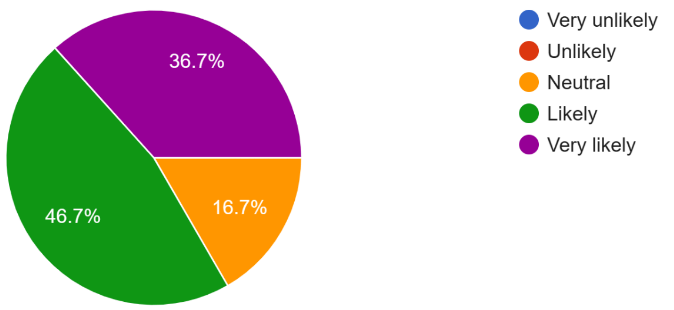
5. How satisfied are you with the amount of information available to you while waiting at a bus stop (e.g. arrival times, seat availability)? (1 = Very Dissatisfied, 5 = Very Satisfied)

30 responses



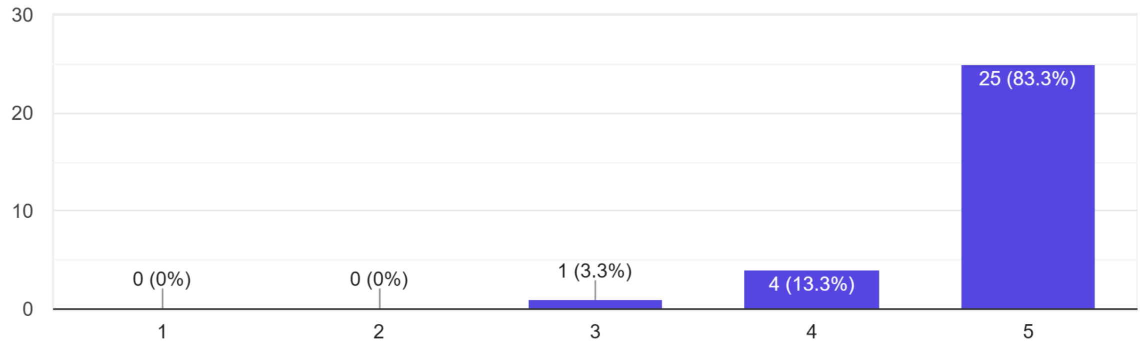
6. If a campus carpool platform existed, how likely would you be to use it?

30 responses



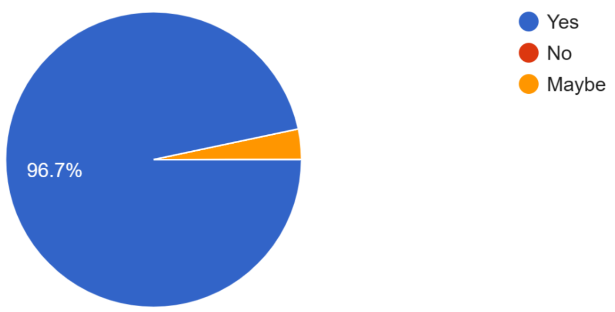
7. How concerned are you about the safety and identity of other users if a carpool feature were available on campus?

30 responses



8. Would you feel more comfortable using a carpool feature if it was restricted to verified UTM student accounts only?

30 responses



9. In your opinion, what is the single biggest problem with UTM's current campus transport system, and how do you think it should be addressed?

30 responses

Response summary

- **Lack of Real-Time Information:** The single biggest problem is the total lack of real-time arrival and location information, which makes planning impossible, causes students to waste time waiting blindly, and results in unreliable schedules.
- **Overcrowding and Capacity Issues:** Buses are frequently too full to board, especially during peak hours, leading to frustration when students wait only to be passed by a packed bus.
- **Proposed Solution (Tracking/App):** The problem should be addressed by modernizing the system with a digital platform or app that provides live tracking, real-time arrival estimates, and, specifically, tracks and shares data on bus capacity/fullness.
- **Bus Delays and Unpredictability:** Bus delays are common and are seriously affecting class times due to the unpredictability of when a bus will arrive.

10. Is there any feature you would like to see in a UTM transport app?

30 responses

Response summary

- **Real-time Location and Tracking:** The most frequently requested features are live location tracking of buses on a campus map and accurate Estimated Time of Arrival (ETA) information, with multiple respondents emphasizing the desire for high reliability and precision.
- **Occupancy and Crowding Information:** Respondents commonly requested features that indicate real-time bus occupancy levels, often suggesting indicators like a "crowd meter" or color-coded systems (green, yellow, red) to show if the bus is empty, half-full, or packed.
- **Additional Functionality:** Other requested features included free carpool options and simple notifications regarding bus arrival information.

JAD session:

Github link: https://github.com/Dc-by1/Group2_Project1_SAD_20252026/tree/main